A sharp norm bound for weighted up-operators on the Boolean lattice

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Abstract

For $a = (a_1, \dots, a_m) \in \mathbb{R}^m$ and $1 \le k \le m$, let $U_k(a) : \ell^2\binom{[m]}{k} \to \ell^2\binom{[m]}{k-1}$ be the weighted up-operator (the weighted $(k, k-1)$ inclusion matrix)

$$(U_k(a)f)(S) = \sum_{i \notin S} a_i \, f(S \cup \{i\}), \qquad |S| = k-1.$$

We prove the sharp inequality $\|U_k(a)\|^2 \le \frac{k(m-k+1)}{m} \sum_{i=1}^m a_i^2$, with equality for all a at the endpoints $k = 1, m$ and, for interior k, if and only if all $|a_i|$ are equal. The classical equal-weight identity $\|U_k(\mathbf{1})\|^2 = k(m-k+1)$ is the case $a = \mathbf{1}$; the weighted statement is not accessible to Johnson-scheme spectral methods, which fail once the weights are generic. The proof is a coordinate-deletion induction whose scalar closure defect factors as a perfect square.

Keywords: Boolean lattice, inclusion matrix, operator norm, Johnson scheme, perfect-square factorization.
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1 Introduction and statement of the result

Let $m \ge 2$ and $1 \le k \le m$. Write $[m] = \{1, \dots, m\}$, let $\binom{[m]}{j}$ denote the j-subsets of $[m]$, and let $\ell^2\binom{[m]}{j}$ be the Euclidean space with orthonormal basis $\{e_S : |S| = j\}$. For weights $a = (a_1, \dots, a_m) \in \mathbb{R}^m$ define the *weighted up-operator* $U_k(a) : \ell^2\binom{[m]}{k} \to \ell^2\binom{[m]}{k-1}$ by

$$(U_k(a)f)(S) = \sum_{i \notin S} a_i f(S \cup \{i\}), \qquad |S| = k-1.$$

In the subset basis $U_k(a)$ is the weighted $(k, k-1)$ inclusion matrix: its (S, T) entry is a_i if $T = S \cup \{i\}$ and 0 otherwise. It is the weighted analogue of the ordinary inclusion (up) operator of the Boolean lattice, whose spectral theory is classical [1, 2, 11]; in particular

$$\|U_k(\mathbf{1})\|^2 = k(m-k+1). \tag{1}$$

A natural question is whether coordinate weights can beat the equal-weight spectral value: for which C does $\|U_k(a)\|^2 \le C \sum_i a_i^2$ hold for all a? The answer is that the rescaled classical constant is already optimal at every level.

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Theorem 1. *For all $m \geq 2$, $1 \leq k \leq m$, and $a \in \mathbb{R}^m$,*

$$\|U_k(a)\|^2 \;\leq\; \frac{k(m-k+1)}{m} \sum_{i=1}^{m} a_i^2.$$

The constant $C(m,k) = \frac{k(m-k+1)}{m}$ is best possible. Moreover:

(i) for $k = 1$ and $k = m$ equality holds for every a;

(ii) for $1 < k < m$ and $a \neq 0$, equality holds if and only if $|a_1| = \cdots = |a_m|$.

The equal-weight case gives $\|U_k(\mathbf{1})\|^2 = k(m-k+1) = C(m,k) \cdot m$, so the theorem asserts that arbitrary coordinate weights, measured in ℓ^2, can never beat the rescaled classical constant, and that among nonzero weights the *only* extremizers in the interior are the equal-magnitude ones. That a proof beyond spectral methods is *necessary* is not an accident: for generic a the family $\{U_k(a)\}$ no longer generates a Bose–Mesner algebra / $\mathfrak{sl}_2$ action commuting with the Johnson scheme, so the standard association-scheme diagonalization is unavailable (cf. Remark 2).

The proof occupies §4: a sign reduction (§3.1), a coordinate-deletion block decomposition (§3.2), a block-norm comparison lemma (§3.3), and a scalar one-variable closure inequality whose defect factors as a perfect square (§4.2). The equality classification is proved in §5 by tracing equality through every step of the induction. §7 records consequences, limitations, and open questions; Appendix A describes the machine-checked identities supporting the proof.

2 Related work

Equal-weight spectra. The spectrum of the unweighted Boolean up operators, and hence (1), is classical; see [1] and the association-scheme diagonalization of [2]. Our theorem contains (1) as the ray $a = \mathbf{1}$ and does not claim it.

Inclusion matrices. Wilson's theorem on inclusion matrices over fields of characteristic zero determines the *rank* of unweighted inclusion matrices (see [9] and the encyclopedic treatment of Johnson-scheme spectra in [11]); it is a rank statement, not a norm statement, and it does not carry weights. Singular-value information for weighted inclusion matrices does not appear to be tabulated; the present result gives the top singular value exactly for the Boolean level pairs.

Differential posets and $\mathfrak{sl}_2$ calculi. Stanley's differential posets [4] provide weighted up/down calculus for enumeration identities, not sharp ℓ^2 operator norms; the $\mathfrak{sl}_2$-type relations behind (1) acquire non-scalar defect terms for generic a (Remark 2).

Analysis of Boolean functions. Reverse hypercontractivity on the cube [3], in the lineage of Bonami–Beckner [7, 8], controls L^p–L^q norms of low-degree polynomials but does not bound a single weighted level-lowering operator in operator norm; the full-space operator $\sum_i a_i \sigma_i^-$ on $\mathbb{R}^{2^{[m]}}$ has norm $\max_k \|U_k(a)\|$, which loses the level-dependent constant. Linear polarization constants of Hilbert spaces [5, 6] concern products of linear functionals and do not imply Theorem 1.

3 Preliminaries

Throughout, $\|\cdot\|$ denotes the operator norm for the Euclidean norms on the spaces involved, and we write $G_k(a) := U_k(a)U_k(a)^{\mathsf{T}}$ on $\ell^2\binom{[m]}{k-1}$, so that $\|U_k(a)\|^2 = \lambda_{\max}(G_k(a))$.

3.1 Sign and phase reduction

We record the reduction in its exact two-sided form, which also covers complex weights. Write $a_i = |a_i| e^{i\theta_i}$ ($\theta_i \in \{0, \pi\}$ for real signs) and define diagonal unitaries D_r on $\ell^2\binom{[m]}{k-1}$ and D_c on $\ell^2\binom{[m]}{k}$ by $(D_r e)_S = e^{-i\sum_{j\in S}\theta_j} e_S$ and $(D_c e)_T = e^{i\sum_{j\in T}\theta_j} e_T$. For $T = S \cup \{i\}$ the (S,T) entry of $D_r\, U_k(|a|)\, D_c$ is $|a_i|\, e^{-i\sum_{j\in S}\theta_j} e^{i\sum_{j\in T}\theta_j} = |a_i| e^{i\theta_i} = a_i$, so

$$U_k(a) \;=\; D_r\, U_k(|a|)\, D_c, \tag{2}$$

a two-sided unitary equivalence; hence $\|U_k(a)\| = \|U_k(|a|)\|$ and the equality classification inherits the reduction. We assume $a \geq 0$ from now on.

3.2 Block decomposition by deleting a coordinate

Write $a = (a', a_m)$ with $a' \in \mathbb{R}^{m-1}$. Order the rows by $(k-1)$-sets not containing m first, then those containing m, and the columns likewise. Since the (S,T) entry is nonzero only when $T = S \cup \{i\}$,

$$U_k(a) \;\simeq\; \begin{pmatrix} U_k(a') & a_m I \\ 0 & U_{k-1}(a') \end{pmatrix}, \tag{3}$$

where I is the $\binom{m-1}{k-1} \times \binom{m-1}{k-1}$ identity indexed by the bijection $S \leftrightarrow S \cup \{m\}$, and $\simeq$ denotes equality after the stated ordering. Multiplying the block forms, and writing $A = U_k(a')$, $B = U_{k-1}(a')$ so that $U_k(a) \simeq \left(\begin{smallmatrix} A & a_m I \\ 0 & B \end{smallmatrix}\right)$ and $U_k(a)^{\mathsf{T}} \simeq \left(\begin{smallmatrix} A^{\mathsf{T}} & 0 \\ a_m I & B^{\mathsf{T}} \end{smallmatrix}\right)$,

$$U_k(a)U_k(a)^{\mathsf{T}} \;\simeq\; \begin{pmatrix} U_k(a')U_k(a')^{\mathsf{T}} + a_m^2 I & a_m\, U_{k-1}(a')^{\mathsf{T}} \\ a_m\, U_{k-1}(a') & U_{k-1}(a')U_{k-1}(a')^{\mathsf{T}} \end{pmatrix}; \tag{4}$$

the off-diagonal block is $a_m\, U_{k-1}(a')^{\mathsf{T}}$ (shape $\binom{m-1}{k-1} \times \binom{m-1}{k-2}$), so its largest singular value is *exactly* $a_m\, \|U_{k-1}(a')\|$.

3.3 A block-norm lemma

Lemma 1. *If $M = \left(\begin{smallmatrix} P & Q \\ Q^{\mathsf{T}} & R \end{smallmatrix}\right)$ is real symmetric with P, R square symmetric, then*

$$\lambda_{\max}(M) \;\leq\; \lambda_{\max} \begin{pmatrix} \lambda_{\max}(P) & s_1(Q) \\ s_1(Q) & \lambda_{\max}(R) \end{pmatrix},$$

where s_1 is the largest singular value.

Proof. For a unit vector $x = (u, v)$, writing $\alpha = |u|$, $\beta' = |v|$,

$$x^{\mathsf{T}} M x \leq \lambda_{\max}(P)\, \alpha^2 + \lambda_{\max}(R)\, \beta'^2 + 2\, s_1(Q)\, \alpha\beta' = y^{\mathsf{T}} S y, \qquad y = (\alpha, \beta'),\ |y| = 1,$$

by Cauchy–Schwarz applied to $u^{\mathsf{T}} Q v$. Maximizing over unit y gives $\lambda_{\max}(S)$. □

4 Proof of Theorem 1

4.1 The induction

We prove the theorem, simultaneously in k, by induction on m, carrying the equality clause. *Induction hypothesis* IH(m') for $m' < m$: the theorem holds for all $1 \le k' \le m'$ and all weights in $\mathbb{R}^{m'}$. The base $m = 2$ is the endpoint case below. The endpoints for general m are exact: $U_1(a)$ is the row vector $(a_1, \dots, a_m)$ and $U_m(a)$ a column vector, so $\|U_1(a)\|^2 = \|U_m(a)\|^2 = |a|^2 = C(m,1)|a|^2 = C(m,m)|a|^2$, and every a is an extremizer, proving (i).

Let $1 < k < m$ and $a \neq 0$. If $a' = 0$ then $U_k(a) = a_m$ times a partial isometry given by the bijection, so $\|U_k(a)\|^2 = a_m^2 < C(m,k)\, a_m^2$ because $k(m-k+1) - m = (k-1)(m-k) > 0$; this also shows no extremizer has exactly one nonzero coordinate. Assume then $s := |a'|^2 > 0$ and set $t = a_m^2/s$, $\alpha^2 = \|U_k(a')\|^2$, $\beta^2 = \|U_{k-1}(a')\|^2$. Applying Lemma 1 to (4), with the exact identity $s_1\big(a_m U_{k-1}(a')^{\mathsf{T}}\big) = a_m\beta$,

$$\|U_k(a)\|^2 \;\le\; \lambda_{\max}\begin{pmatrix} \alpha^2 + a_m^2 & a_m\beta \\ a_m\beta & \beta^2 \end{pmatrix}.$$

By IH($m-1$), $\alpha^2 \le A\,s$ and $\beta^2 \le B\,s$ with $A = C(m-1,k) = \frac{k(m-k)}{m-1}$ and $B = C(m-1,k-1) = \frac{(k-1)(m-k+1)}{m-1}$; both children lie in range because $1 \le k-1$ and $k \le m-1$. Since $\lambda_{\max}\left(\begin{smallmatrix} p & o \\ o & q \end{smallmatrix}\right) = \frac{1}{2}\big(p + q + \sqrt{(p-q)^2 + 4o^2}\big)$ is nondecreasing in each of p, q and in $|o|$ (the partial derivatives have the form $\frac{1}{2}[1 \pm (p-q)/\sqrt{(p-q)^2+4o^2}] \ge 0$, and the bound in o is even),

$$\|U_k(a)\|^2 \;\le\; s \cdot \lambda_{\max}\begin{pmatrix} A + t & \sqrt{tB} \\ \sqrt{tB} & B \end{pmatrix}. \tag{5}$$

4.2 Scalar closure

Lemma 2. *For $m \ge 3$, $1 \le k \le m$, $t \ge 0$, with A, B as above and $C = C(m,k) = \frac{k(m-k+1)}{m}$,*

$$\lambda_{\max}\begin{pmatrix} A + t & \sqrt{tB} \\ \sqrt{tB} & B \end{pmatrix} \;\le\; C(1+t),$$

and for interior k equality holds if and only if $t = \frac{1}{m-1}$.

Proof. The largest eigenvalue equals $\frac{1}{2}\big(A + B + t + \sqrt{(A-B+t)^2 + 4Bt}\big)$, so the claim is $\sqrt{(A-B+t)^2 + 4Bt} \le X(t)$ with $X(t) = 2C(1+t) - A - B - t$. Now X is affine with slope $2C - 1 = \frac{2k(m-k+1)-m}{m} > 0$ and intercept $X(0) = 2C - A - B = \frac{k(k-1)+(m-k)(m-k+1)}{m(m-1)} > 0$; hence $X(t) > 0$ for $t \ge 0$ and squaring is legitimate. The squared defect factors exactly:

$$X(t)^2 - \big[(A-B+t)^2 + 4Bt\big] = \frac{4k(k-1)(m-k)(m-k+1)}{m^2(m-1)^2}\,\big((m-1)t - 1\big)^2 \;\ge\; 0,$$

a polynomial identity in m, k, t verified symbolically (see Appendix A). For interior k the prefactor is positive, so equality holds iff $t = \frac{1}{m-1}$. □

Combining (5) with Lemma 2, $\|U_k(a)\|^2 \le s\,C(1+t) = C(m,k)(s + a_m^2) = C(m,k)|a|^2$, completing the induction. Sharpness: $G_k(\mathbf{1})\mathbf{1} = k(m-k+1)\mathbf{1}$, so equality is attained at $a = \mathbf{1}$ in every level and no smaller constant works at any level. □

Corollary 1. *Theorem 1 remains true for $a \in \mathbb{C}^m$ with $\sum_i |a_i|^2$ on the right, and the equality classification is unchanged with $|a_i|$ in place of a_i.*

Proof. By (2), $\|U_k(a)\| = \|U_k(|a|)\|$ entrywise in magnitudes, and $\sum_i a_i^2$ is replaced by $\sum_i |a_i|^2$ throughout; equality holds for a exactly when it holds for $|a|$. □

5 Equality cases

Assume $1 < k < m$ and equality in Theorem 1 with $a \neq 0$. Tracing the chain in §4: Lemma 2 must be an equality, forcing $t = \frac{1}{m-1}$; since then $t > 0$ and $B > 0$ the corner off-diagonal $\sqrt{tB}\, s$ is nonzero, and the monotonicity step (5) is strict in each diagonal entry when the off-diagonal is nonzero (the partial derivatives of $\lambda_{\max}$ are $\frac{1}{2}\big[1 \pm \frac{p-q}{\sqrt{(p-q)^2+4o^2}}\big] > 0$ for $o > 0$, because $\sqrt{(p-q)^2+4o^2} > |p-q|$), forcing $\alpha^2 = As$ and $\beta^2 = Bs$. Thus equality holds at the subproblem $(m-1,k)$ on a' (and at $(m-1,k-1)$). For $m \geq 4$ at least one of these subproblems is interior — if $2 \leq k \leq m-2$ then $(m-1,k)$ is; if $k = 2$ (resp. $k = m-1$) then $(m-1,2)$ (resp. $(m-1,m-2)$) is interior for $m \geq 4$ — and the strengthened induction hypothesis yields $|a_1| = \cdots = |a_{m-1}|$; then $t = \frac{1}{m-1}$ gives $a_m^2 = s/(m-1)$, the same common value. The remaining base $m = 3, k = 2$ is computed exactly: with $x_i = a_i^2$ and $s = x_1 + x_2 + x_3$, the matrix $\frac{4}{3}s\, I - G_2(a)$ has principal minors $\frac{x_1+x_2+4x_3}{3}$, $\frac{s(x_1+4x_2+4x_3)}{9}$; these are the $(3,3)$ entry and the 2×2 principal minor on indices $\{2,3\}$ in the basis ordering $e_{\{1\}}, e_{\{2\}}, e_{\{3\}}$, and the remaining principal minors are obtained by permuting the coordinate labels $1,2,3$, and determinant $\frac{4}{27}\big[(x_1+x_2+x_3)^3 - 27x_1x_2x_3\big] \geq 0$ by AM–GM, with equality iff $x_1 = x_2 = x_3$. Hence $\lambda_{\max}(G_2(a)) = \frac{4}{3}s$ iff $a_1^2 = a_2^2 = a_3^2$. The converse direction (equal magnitudes give equality) follows from the sign reduction (2) and $G_k(\mathbf{1})\mathbf{1} = k(m-k+1)\mathbf{1}$. This completes the proof of Theorem 1(ii). □

Remark 1 (complement duality). Let $C_j : \ell^2\binom{[m]}{j} \to \ell^2\binom{[m]}{m-j}$, $e_S \mapsto e_{S^{\mathsf{c}}}$. Since $(S \cup \{i\})^{\mathsf{c}} = S^{\mathsf{c}} \setminus \{i\}$, entrywise comparison gives $C_{k-1}\, U_k(a)\, C_k = U_{m-k+1}(a)^{\mathsf{T}}$, hence $\|U_k(a)\| = \|U_{m-k+1}(a)\|$ and $C(m,k) = C(m,m-k+1)$; the induction above therefore needs only the range $k \leq \frac{m+1}{2}$.

Remark 2 (spectral methods fail for generic weights). For $a = \mathbf{1}$ the operators $\{U_k, U_k^{\mathsf{T}}\}$ act between the homogeneous components of the Boolean lattice with $[D,U] = U$-type relations after weighting by rank, and the spectra follow from the Johnson scheme [1]. For generic a the commutation relations acquire non-scalar defect terms and no association scheme survives; the deletion induction appears to be the only route to a sharp general result.

6 Application to a squarefree arithmetic compression

We illustrate Theorem 1 with an operator arising in analytic number theory. The application is deliberately restricted: the weighted bound controls the *compression* of the arithmetic operator to the squarefree indices, and nothing in this section extends that control to the full operator.

6.1 The arithmetic operator and its normalization

Fix an integer $N \geq 2$ and write $\mathcal{P} = \{p \leq N : p \text{ prime}\}$, $m = \pi(N)$, with $\mathcal{P} = \{p_1 < \cdots < p_m\}$. Let φ denote Euler's function and let $\ell^2(\varphi)$ be $\mathbb{C}^{\{1,\ldots,N\}}$ with norm $\|y\|_\varphi^2 = \sum_{r \leq N} \varphi(r)\, |y_r|^2$. Define

$$(T_N y)(r) = \sum_{k \leq N/r} \Lambda(k)\, y_{rk}, \qquad 1 \leq r \leq N, \tag{6}$$

where Λ is the von Mangoldt function. Multiplication by $\sqrt{\varphi}$ is a unitary identification of $\ell^2(\varphi)$ with ordinary ℓ^2, under which T_N becomes the matrix

$$\widetilde{T}_N[r,s] = \begin{cases} \Lambda(s/r)\,\sqrt{\varphi(r)/\varphi(s)}, & r \mid s,\ s \neq r, \\ 0, & \text{otherwise}, \end{cases} \tag{7}$$

so that $\|T_N\|_{\ell^2(\varphi)} = \|\widetilde{T}_N\|_{\ell^2}$ exactly.

Let P_{sqf} be the orthogonal projection of ℓ^2 onto the span of the squarefree indices. Squarefree integers whose prime factors lie in $\mathcal{P}$ are in bijection with subsets $S \subseteq [m]$ via $r \leftrightarrow \{i : p_i \mid r\}$; write H_j for the span of the squarefree indices with exactly j prime factors, so that $P_{\mathrm{sqf}}\ell^2 = \bigoplus_{j\geq 0} H_j$.

6.2 The compression identity

Call $S \subseteq [m]$ admissible if $\prod_{i\in S} p_i \leq N$; the admissible sets form a down-set (an order ideal), and H_j is non-zero for $j \leq k_{\max}(N) := \max\{k : p_1 \cdots p_k \leq N\}$ and zero beyond it. For weights $a_i = \log p_i/\sqrt{p_i - 1}$, let V_k denote the compression of the operator $U_k(a)$ of Theorem 1 to the admissible levels: for $|S| = k-1$ admissible,

$$(V_k f)(S) = \sum_{\substack{i\notin S \\ \prod_{j\in S} p_j \cdot p_i \leq N}} a_i\, f(S\cup\{i\}), \qquad |S| = k-1,$$

extended to all of H_k by $V_k f = V_k P_{\mathrm{adm},k} f$, where $P_{\mathrm{adm},k}$ is the orthogonal projection of H_k onto the span of the admissible k-sets.

Lemma 3 (compression identity)**.** *With the notation above,*

$$P_{\mathrm{sqf}}\,\widetilde{T}_N\,P_{\mathrm{sqf}} \;=\; \sum_{k=1}^{k_{\max}(N)} V_k \qquad \textit{(entrywise, exactly).}$$

Proof. An entry $\widetilde{T}_N[r,s]$ with $r \mid s$ and $s \neq r$ is non-zero only when s/r is a prime power p^j, since Λ vanishes elsewhere. If r and s are both squarefree and $s = r\,p^j$ with $j \geq 2$, then $p^2 \mid s$, a contradiction; hence $j = 1$ and $s = rp$ with $p \nmid r$. For such a pair, $\varphi(rp) = \varphi(r)(p-1)$, so the entry equals $\log p/\sqrt{p-1} = a_p$: the weight of the unique edge $S \to S\cup\{i\}$ of the Boolean lattice. All other squarefree pairs give a zero entry. The survival condition is exactly $s = rp \leq N$, i.e. admissibility of $S\cup\{i\}$, and the level shifts by one. This is entrywise the matrix of $\sum_k V_k$ under the subset parametrization. □

6.3 The norm estimate

Theorem 2 (squarefree compression bound)**.** *For every $N \geq 2$,*

$$\left\|P_{\mathrm{sqf}}\,T_N\,P_{\mathrm{sqf}}\right\|^2_{\ell^2(\varphi)} \;=\; \sup_{1\leq k\leq k_{\max}(N)} \|V_k\|^2 \;\leq\; k_{\max}(N) \sum_{p\leq N} \frac{(\log p)^2}{p-1}.$$

Consequently, unconditionally,

$$\left\|P_{\mathrm{sqf}}\,T_N\,P_{\mathrm{sqf}}\right\|^2 \;\leq\; \left(\tfrac{1}{2} + o(1)\right) \frac{(\log N)^3}{\log\log N} \qquad (N \to \infty).$$

Proof. Each V_k maps H_k into H_{k-1} and vanishes on the other levels: $V_k = P_{k-1}V_kP_k$ (support restriction in the domain, range in H_{k-1}), hence $V_k^* = P_kV_k^*P_{k-1}$. Therefore for $k \neq j$, $V_k^*V_j = P_kV_k^*P_{k-1}P_{j-1}V_j = 0$. Hence, writing $S := \sum_k V_k$,

$$S^*S = \sum_k V_k^*V_k, \qquad \langle S^*Sx, x\rangle = \sum_k \|V_kx_k\|^2 \le \Big(\sup_k \|V_k\|^2\Big) \sum_k \|x_k\|^2$$

for $x = \bigoplus_k x_k \in P_{\mathrm{sqf}}\ell^2$, and the reverse inequality holds by taking x in a single level. Thus $\|S\|^2 = \sup_k \|V_k\|^2$; this is an equality, not a triangle-inequality accumulation. Compression to the admissible levels cannot increase the norm, so $\|V_k\| \le \|U_k(a)\|$, and Theorem 1 gives

$$\|V_k\|^2 \le \frac{k(m-k+1)}{m} \sum_{i=1}^{m} a_i^2 \le k \sum_{p\le N} \frac{(\log p)^2}{p-1} \le k_{\max}(N) \sum_{p\le N} \frac{(\log p)^2}{p-1}.$$

For the asymptotic form we use two standard consequences of the prime number theorem (partial summation; see Montgomery–Vaughan [10]). First, $\vartheta(x) = x+o(x)$ gives $\log(p_1\cdots p_k) = \vartheta(p_k) \sim k\log k$, so the largest achievable level satisfies $k_{\max}(N) = (1+o(1))\log N/\log\log N$. Second, $\vartheta(x) = x + o(x)$ gives $\sum_{p\le x}(\log p)/p = \log x + o(\log x)$, and partial summation then yields

$$\sum_{p\le x} \frac{(\log p)^2}{p} = (\log x)^2 - \int_2^x \frac{\log t + o(\log t)}{t}\, dt = \tfrac{1}{2}(\log x)^2 + o\big((\log x)^2\big);$$

replacing $1/p$ by $1/(p-1)$ changes the sum by $O(1)$: for $p \ge 2$, $(\log p)^2/(p(p-1)) \le 2(\log p)^2/p^2$, and $\sum_p (\log p)^2/p^2 \le \sum_{n\ge 2}(\log n)^2/n^2 < \infty$. Multiplying the two asymptotics gives the claim. □

Remark 3 (level-one sharpness and scope). V_1 is a row operator and $\|V_1\|^2 = \sum_{p\le N} a_p^2 = \big(\frac{1}{2} + o(1)\big)(\log N)^2$, so the level-$k$ use of Theorem 1 is asymptotically sharp at $k = 1$, and $\|P_{\mathrm{sqf}}T_NP_{\mathrm{sqf}}\| \ge \|V_1\|$ already accounts for a $(\log N)^2$ lower growth of the squared norm. Numerical experiments (see the repository's `verification/` directory) indicate that the compression norm is attained at e_1 for small N, suggesting the true order of $\|P_{\mathrm{sqf}}T_NP_{\mathrm{sqf}}\|^2$ may be $\Theta\big((\log N)^2\big)$; this is not proved here, and the factor $k_{\max}(N)$ in the upper bound should be read as an artifact of controlling every level by the worst level.

Remark 4 (what this result does not say). Theorem 2 bounds the compression $P_{\mathrm{sqf}}T_NP_{\mathrm{sqf}}$ only. Entries of $\widetilde{T}_N$ with a non-squarefree index survive in the two off-diagonal corners and in the nonsquarefree block; they are not Boolean-lattice edges and are not controlled by Theorem 1. In particular, nothing here implies $\|T_N\| = O(\log N)$ for the full operator, nor does it address Nyman–Beurling distances, zero-free regions for ζ, or the Riemann hypothesis. For orientation we record one rigorous complement: since the operator lowers indices, $T_N^*e_1$ (not T_Ne_1, which vanishes) satisfies

$$\|T_N\|^2 \;\ge\; \|T_N^*e_1\|_\varphi^2 = \sum_{2\le k\le N} \frac{\Lambda(k)^2}{\varphi(k)} \;\ge\; \sum_{p\le N} \frac{(\log p)^2}{p-1} = \Big(\tfrac{1}{2} + o(1)\Big)(\log N)^2,$$

so $\|T_N\| \ge (\sqrt{2}^{-1} + o(1))\log N$: a lower bound, in the opposite direction from the compression estimate.

7 Concluding remarks

Consequences. Theorem 1 identifies the exact worst case of a weighted level-lowering operator on the Boolean lattice: the worst weights are the flat ones, up to signs and phases (Corollary 1), and the extremal rigidity is exact rather than asymptotic. By complement duality (Remark 1) the same bound and classification hold for the weighted down-operators $U_k(a)^{\mathsf{T}}$. The scalar closure lemma is an independent one-variable inequality that may be reusable wherever a 2×2 corner comparison of this shape arises.

Limitations. The method is specific to the Boolean lattice level structure: the deletion recursion produces exactly two children with the same ambient parameters, and the perfect-square factorization depends on the constants $C(m, k)$. No comparison with the weighted spectra of $G_k(a)$ beyond the top eigenvalue is obtained, and nothing in this paper addresses the infinite-dimensional operator-theoretic problems (e.g. Nyman–Beurling-type formulations) that motivated the search; any connection there is conditional and unproved.

Open questions. (1) For which graded posets with rank numbers analogous to $\binom{mj}{j}$ does the deletion induction close with a perfect-square defect? (2) Determine $\lambda_{\max}(G_k(a))$ exactly as a function of the sorted magnitudes of a, not only its maximum over the sphere $\{|a|^2 = 1\}$. (3) Analogous sharp bounds for the two-step operators $U_k(a)U_{k+1}(a)$, where the corner comparison no longer closes at one variable.

References

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A Reproducibility of the checked identities

The proof above contains no step that requires computer assistance; the machine checks below are regression evidence, not part of the proof.

Closure identity. With A, B, C as in Lemma 2, the defect factorization

$$X(t)^2 - \big[(A - B + t)^2 + 4Bt\big] = \frac{4k(k-1)(m-k)(m-k+1)}{m^2(m-1)^2}\big((m-1)t - 1\big)^2$$

was verified by expansion and simplification in `sympy` as a rational-function identity in *general* symbols (m, k, t), not by sampling.

Independent reproduction. A verification script built from the definition of $U_k(a)$ alone (no reuse of the proof) checks the block identity (3), the Gram identity (4), the duality identity of Remark 1, and the phase identity (2) entrywise for small parameters, and evaluates the inequality at 30 exact rational instances (m, k, a) (worst observed ratio 0.979016, zero violations), using certified real-root isolation for the spectral maxima. Exact checks of $\lambda_{\max}(U_k(\mathbf{1})U_k(\mathbf{1})^{\mathsf{T}}) = k(m-k+1)$ were run for $2 \le m \le 8$, all k.

Artifacts. The scripts producing the checks described above, with pinned dependency versions and reproduction instructions, are provided in the `verification/` directory of the accompanying repository (`weighted-johnson-inequality`); the symbolic script re-derives the closure identity in general symbols and the exact script rebuilds $U_k(a)$ from its definition alone. No mathematical step of this manuscript depends on any of these machine checks.